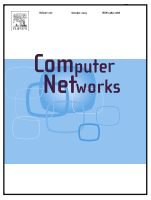
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NOMA-based energy efficiency optimization for UAV-assisted relay communication networks

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ABSTRACT

Unmanned aerial vehicles (UAVs) are often employed to wireless networks as mobile relays to extend network coverage or enhance network performance for emergencies. In these applications, the UAV may serve a large number of ground users with few resources and the users may have various requirements for the quality of service (QoS). To tackle these challenges, this paper investigates different communication resources and diverse QoS requirements of users, and proposes a UAV-assisted real-time relay communication scheme based on NOMA, where a UAV as the aerial relay communicates with multiple users simultaneously. In order to prolong the network lifetime, we formulate the energy efficiency maximization problem by jointly optimizing user scheduling, quality of service (QoS), transmission power, and UAV's flight trajectory. Due to the non-convex, mixed-integer nature of the optimization problem caused by coupling among multiple variables and NOMA decoding rules, we decompose it into three subproblems: user scheduling, transmission power allocation and UAV trajectory planning. The user scheduling subproblem is solved by using an integer programming approach. The transmission power allocation is addressed through a combination of successive convex approximation (SCA) technique and convex optimization method. For the UAV trajectory planning, a suitable solution is obtained by applying SCA and convex optimization in conjunction with the Dinkelbach transformation and the introduction of auxiliary slack variables. Then the three subproblems are iteratively optimized by the block coordinate descent (BCD) method to maximize system energy efficiency. Simulation results verify that the proposed scheme outperforms the benchmark approaches in terms of energy efficiency, thereby effectively extending network lifetime.

1. Introduction

With the rapid development of digitization and informatization, wireless communication networks have become indispensable infrastructures in modern society for various applications, such as transportation, logistics, smart cities, agriculture, industrial internet, etc[1–3]. However, natural disasters (e.g., earthquakes, floods) or accidents may damage base stations or ground equipment, resulting in sudden communication interruptions. This poses severe challenges to network connectivity and information dissemination. To tackle this, employing unmanned aerial vehicle (UAV) as mobile platform to establish temporary real-time relay communication networks is an effective solution. Since UAV leverages high mobility and good line-of-sight (LoS) links [4–6], it can establish temporary relay networks quickly and restore communication connectivity for emergency or disaster scenarios [7–10].

Although UAVs exhibit significant potential in rapidly and conveniently establishing relay communication networks in emergency situations, their limited onboard battery capacity severely restricts the duration of the network's operation. Thus, in UAV-assisted relay communication networks, optimizing the energy efficiency, which is typically defined as the ratio of network throughput to energy consumption, has become a critical issue. Most of existing studies on energy efficiency optimization mainly follow two directions. One is to minimize the network energy consumption while maintaining the same throughput [11–14], and the other is to maximize the system throughput while keeping the energy consumption constant [15–20].

To avoid interference among multiple users, Orthogonal Multiple Access (OMA) has been often adopted in UAV-assisted wireless communication networks. OMA allocates separate frequencies or resource blocks or time slots or codes to the different users at the same time in the same cell while accessing the network resources to eliminate interference among users. For high-density and large-capacity communication scenarios, Non-Orthogonal Multiple Access (NOMA) is introduced into wireless networks to improve system spectral efficiency, thereby

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Table 1
Comparison with existing schemes..

Reference	[12]	[15]	[20]	[25]	[28]	[29]	[30]	[31]	[32]	[33]	Ou work
NOMA				✓	✓			✓	✓		✓
Various Quality of service		✓			✓						✓
UAV trajectory planning	✓		✓			✓	✓	✓	✓	✓	✓
User scheduling	✓		✓					✓	✓		✓
Transmission power allocation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

increasing network capacity. Compared to OMA, NOMA employs power domain multiplexing or code domain multiplexing and permits multiple users to share the same time-frequency resource blocks. This theoretically offers higher spectral efficiency and the capability to serve a larger number of users simultaneously, making it a pivotal technology for enhancing network capacity within the same frequency spectrum [21,22]. Many researches have applied NOMA to UAV-assisted relay communication networks to improve network communication performance [23–28].

In UAV-assisted wireless communication networks, due to the high mobility, the flight attitude and trajectory of UAV have a significant impact not only on the network performance but also on the energy consumption of the system. For instance, when the UAV gets closer to the ground users, the transmission rate between UAV and the users may be higher, resulting in a greater network throughput, but it may consume more energy. Therefore, several studies have delved into the energy efficiency optimization for UAV-assisted relay communication networks by jointly optimizing flight trajectory of UAV and the allocation of various communication resources [29–33]. The comparison of existing UAV-assisted relay communication schemes on energy efficiency optimization is listed in Table 1.

Despite these advances, practical UAV-assisted relay networks in emergency scenarios still face some challenges. Firstly, UAV typically serves multiple users with varying QoS requirements. For example, in a search and rescue scenario, the users in key areas require higher transmission rates and better communication service. Secondly, in emergency situations, the communication resources, such as time-frequency allocation, are often scarce and it is difficult to meet the communication demands of multiple users simultaneously. Thirdly, the restricted cruise time of UAV presents as a bottleneck. So, with limited resources, how to provide communication connections that meet various QoS requirements of different users is worth further discussion. Motivated by these, we investigate a UAV-assisted real-time relay communication network, where the UAV act as an aerial mobile relay, serving multiple users on the ground. Under the constraints of communication resources, we introduce NOMA technology to promote energy efficiency. Then, considering users' QoS requirements and UAV's energy consumption, we jointly optimize user scheduling, power allocation and UAV trajectory to maximize system energy efficiency. The main contributions of this paper are summarized as follows.

- To tackle the communication interruption between the base station and multiple ground users in emergency, we establish a NOMA-based UAV-assisted real-time relay network. Considering limited communication resources, UAV energy constraints, and diverse user QoS requirements, we formulate a comprehensive energy efficiency optimization problem to extend network lifetime.
- Due to the non-convex, mixed-integer nature of the original problem caused by coupling between variables and NOMA decoding rules, we decompose it into three independent sub-problems: user scheduling, transmission power allocation and UAV trajectory planning. Then, the user scheduling is solved via integer programming to select near-end/far-end users in each time slot, ensuring channel quality differentiation for NOMA. The power allocation is addressed by using successive convex approximation (SCA) and convex optimization to handle non-convex rate functions of far-end users. The UAV trajectory planning is optimized via SCA, convex optimization, Dinkelbach transformation, and auxiliary slack variables, while accurately modeling velocity-dependent propulsion energy consumption.
- We adopt the block coordinate descent (BCD) method to iteratively solve the three subproblems, ensuring convergence to a feasible solution for the original energy efficiency optimization problem. Extensive simulations are conducted to analyze the impact of user scheduling, QoS requirements, transmission power, and UAV trajectory on network energy efficiency. And the results verify that the proposed scheme outperforms benchmark approaches in terms of energy efficiency.

The remainder of this paper is organized as follows. Section II describes the system model. Section III formulates the optimization problem. In Section IV, the solution of proposed optimization problem is presented. Simulation results are discussed in Section V, and Section VI summarizes the paper.

2. Related work

In UAV-assisted wireless networks, UAV's endurance is highly dependent on the energy carried by its onboard battery. To prolong the network lifetime, several studies have focused on energy minimization for UAV-assisted communication systems [11–14]. In [11], the authors comprehensively considered propulsion energy of UAV and communication energy in a downlink UAV-aided communication system, provided online learning and timely energy-saving solutions based on deep reinforcement learning (DRL), then proposed an actor-critic-based deep stochastic online scheduling algorithm to minimize network energy[12]. designed a UAV-assisted bidirectional relaying system for multi-pair user devices, and jointly optimized transmission scheduling and association, UAV relay and user device transmission power, and UAV trajectory to minimize the actual overall energy consumption of the UAV relay while ensuring the minimum required transmission rate, thereby maximizing system energy efficiency[13]. developed an energy-efficient UAV-assisted collaborative mobile edge computing (MEC) system, decomposing task offloading and trajectory planning into three convex subproblems to minimize user terminal energy consumption[14]. investigated joint energy and trajectory optimization for UAV-enabled relaying networks with multiple source-destination pairs, and minimized total propulsion and communication energy while satisfying all users' communication requirements.

In addition to energy-saving, enhancing network throughput with energy constraints is an essential way to improve energy efficiency for UAV-assisted communication networks based on OMA technology [15–20][15,16]. studied UAV-enabled full-duplex relay network and optimized design of beamforming and power allocation to maximize the data rate. The authors in [17] maximized the sum user throughput by jointly optimizing the 3-D UAV trajectory and velocity-time profile in the presence of onboard energy, velocity, acceleration, and completion time constraints for a UAV-assisted Internet of Things communication system[18]. investigated a UAV-assisted downlink cellular network, where a UAV served as an aerial base station and powered by a wireless charging station on the ground, and maximized the common throughput for two resource sharing schemes: Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) by jointly optimizing UAV hover positions and communication resource allocation.

In [19], a new Integrated Periodic Sensing and Communication (IPSAC) mechanism for a UAV-enabled integrated sensing and communication system was proposed to maximize system achievable rate while meeting the sensing frequency and beam pattern gain requirement for the given targets. To fully exploit the flying flexibility of UAV in serving multiple source-destination pairs, [20] designed UAV flight trajectories and proposed a scheduling protocol based on TDMA to maximize the system throughput.

OMA puts a stringent constraint on the spectral efficiency as well as the number of users that UAV serves in a particular UAV-assisted communication network. To further enhance the network capacity, NOMA is often applied in UAV-assisted networks supporting a large number of devices with few resources to enhance system performance [23–28]. In [23], the authors developed an unmanned aerial vehicle (UAV)-assisted non-orthogonal multiple access (NOMA) short-packet communication scheme with imperfect interference cancellation, which achieves near-optimal performance under low interference levels and outperforms conventional OMA schemes, thereby attaining superior system performance[24]. proposed a subchannel allocation algorithm based on the classical K-means clustering method and matching theory. By jointly optimizing subchannel allocation, node uplink transmission power and UAV flight altitude, system capacity was maximized[25]. combined NOMA with decode-and-forward, and proposed a joint optimization problem of UAV altitude, channel allocation and power allocation to maximize the total rate of edge users covered by UAV in macro-cell networks. In [26], two optimization objectives were conducted in a NOMA network. The first one was to minimize the transmission power of the UAV subject to the minimum achievable rate requirements, and the second one was to maximize the achievable rate of specific user subject to the minimum achievable rate requirements of other users. In [27] a UAV-assisted NOMA transmission scheme was proposed to achieve secure downlink transmission by artificial jamming. It investigated the trade-off between artificial interference power and communication rate in the presence of passive eavesdroppers, and the goal was to balance security and transmission performance. The authors in [28] proposed a Reconfigurable Intelligent Surface (RIS)-assisted bidirectional full-duplex communication and NOMA transmission scheme, and enhanced spectral efficiency through optimized power allocation, thereby achieving significant system performance improvement.

The energy efficiency of UAV-assisted relay communication networks is influenced by a variety of factors, and they are mutually coupled and intertwined. Therefore, several studies have further focused on jointly optimizing various resources and factors to enhance energy efficiency for UAV-assisted communication networks. In the presence of potential eavesdroppers, [29] maximized the secrecy energy efficiency by jointly optimizing the communication scheduling, power allocation and UAV trajectory for a UAV-enabled mobile relaying network with single user[30]. used fuel-powered UAV to build a relay communication network between a source node and a destination node, and maximized network energy efficiency by designing the UAV's flight trajectory and communication resource allocation[31]. studied the energy efficiency problem of NOMA-based air-ground-space integrated relay systems, and achieved maximal energy efficiency through UAV trajectory design and power allocation. In [32], authors jointly optimized user scheduling, power allocation and UAV trajectory to maximize energy efficiency for UAV-based NOMA downlink wireless networks with the Quality of Service (QoS) requirements. To overcome severe path fading between two users and achieve sufficient signal gain for demodulation, [33] proposed a three-node cooperative communication system using a UAV as a bidirectional relay and obtained the optimal energy efficiency.

3. System model

As shown in Fig. 1, we consider a wireless relay network deployed in an emergency or disaster scenario. It consists of a base station (*BS*), a UAV and M ground users. The ground users are randomly distributed

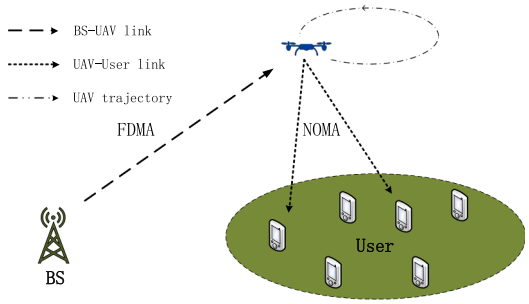


Fig. 1. System model.

in a limited area, and the two-dimensional coordinates of the m-th $m \in \mathcal{M} \triangleq \{1, 2, \ldots, M\}$ user are denoted as $U_m = (x_m, y_m)$. The coordinates of the *BS* are denoted as $BS = (x_{BS}, y_{BS})$. Due to communication link blockage between the *BS* and ground users, a rotary-wing UAV is adopted to establish real-time relay communication links, and it promptly forwards data from the *BS* to the ground users using DF mode. FDMA technology is used for communication between the *BS* and the UAV. To improve network communication capacity, NOMA technology is employed for the downlink transmission between the UAV and ground users.

Assuming the completion time of the UAV mission is T, it is discretized into N sub-slots of continuous duration δ_t, where $\delta_t = T/N$. Each sub-slot is indexed as $n \in \mathcal{N} \triangleq \{1, 2, \ldots, N\}$. Analogous to the approach in [12–15], the UAV flies at a constant altitude H, and $\mathbf{W}[n] = (x[n], y[n])$ is the position of the UAV in the nth sub-slot.

As shown in the 3GPP specification [34], the channel model between the UAV and ground users depends on environmental scenarios, such as the suburban with less scattering and the macro urban with rich scattering. Especially when the UAV flies above $40m$ in the rural area, the UAV offers a nearly 100% LoS probability for UAV-ground channel. Since this study is concerned with relay communication for emergency scenarios in the suburban, and the UAV flies at a sufficiently high altitude, like [5,10,13,14], the channel model between the *BS* and the UAV, as well as between the UAV and the user, adopts LoS links. Therefore, the channel power gain of the *BS*-UAV link in the nth sub-slot is expressed as

$$h_n^{uav} = \frac{b_0}{\|\mathbf{W}[n] - BS\|^2 + H^2} \tag{1}$$

where $\|\cdot\|$ is the Euclidean norm, b_0 is the channel power gain when the distance from the UAV to the *BS* is the reference distance. The channel power gain from the UAV to the mth user in the nth sub-slot can be obtained as

$$h_n^m = \frac{b_0}{\|\mathbf{W}[n] - U_m\|^2 + H^2} \tag{2}$$

4. Problem formulation

4.1. Problem description

The *BS* uses a total bandwidth B and a constant transmission power to simultaneously transmit information to M users via UAV's forwarding. It is assumed that the bandwidth used for transmitting data for each user is equal, i.e. $B_m = B/M$. Therefore, like [9], the data transmission rate of the *BS*-UAV link in the nth sub-slot is written as

$$R_{n_1} = B_m \log_2\left(1 + \frac{P_{BS} h_n^{uav}}{\sigma^2}\right) \tag{3}$$

where P_{BS} is the transmission power of the *BS*, and σ^2 is the noise power.

At the receiving end, each ground user uses the SIC technique to demodulate their data from the UAV [25]. In this paper, it is assumed that each UAV can simultaneously serve two users. From formula (2), it can be seen that the channel quality of users is determined by their distance from the UAV, users farther away have poorer channel quality, while users closer have relatively better channel quality. At the UAV, limited transmission power is required to simultaneously send information to two users, hence need for power allocation. First, define the user service type flag $nf = \{near, far\}$, where $nf = near$ indicates a near-end user, $nf = far$ indicates a far-end user. P_n^{near} and P_n^{far} represent the transmission power allocated to the near-end and far-end users served by the UAV in the nth sub-slot. According to decoding rule of NOMA [31], we have $P_n^{near} < P_n^{far}$.

To indicate whether user m is served in the nth sub-slot, define the user scheduling function $F(n, m, nf) = \{0, 1\}$, where $F(n, m, nf) = 1$ expresses user m being served as an nf-end user by the UAV in the nth sub-slot, and $F(n, m, nf) = 0$ indicates that user m is not served as an nf-end user. Since in one sub-slot, only two of M users are selected, one is served as a near-end user and the other is as far-end user, we have the constraints as

$$\sum_{m=1}^{M} F(n, m, near) = 1, \forall n \tag{4}$$

$$\sum_{m=1}^{M} F(n, m, far) = 1, \forall n \tag{5}$$

The channel quality of the near-end user is better than that of the far-end user, it is written as

$$\sum_{m=1}^{M} \left(F(n, m, near) h_n^m \right) > \sum_{m=2}^{M} \left(F(n, m, far) h_n^m \right), \forall n \tag{6}$$

Assuming that in the nth sub-slot, user m is served as a near-end user or a far-end user, the achievable communication rates in the UAV-user link are represented as [32]

$$R_{n_near_2}^m = B\log_2\left(1 + \frac{h_n^m P_n^{near}}{\sigma^2}\right) \tag{7}$$

$$R_{n_far_2}^m = B\log_2\left(1 + \frac{h_n^m P_n^{far}}{P_n^{near} h_n^m + \sigma^2}\right) \tag{8}$$

As the UAV adopts real-time forwarding for data, the achievable communication rate at the user end is the minimum of the communication rates in the *BS*-UAV link and the UAV-user link. Combining user scheduling, the achievable communication rates for user m as a near-end user or a far-end user in the nth sub-slot are written as

$$R_{n_near}^m = min\left\{ R_{n_1},\ F(n, m, near) R_{n_near_2}^m \right\} \tag{9}$$

$$R_{n_far}^m = min\left\{ R_{n_1},\ F(n, m, far) R_{n_far_2}^m \right\} \tag{10}$$

So, the achievable communication rate for user m in the nth sub-slot can be obtained as

$$R_n^m = R_{n_near}^m + R_{n_far}^m \tag{11}$$

In UAV-assisted relay communication networks, BS and ground users generally have a stable power supply, and their energy consumption hardly affects the duration of the network, while UAVs are usually powered by onboard batteries and their energy cannot be replenished in time. So, the lifetime of UAV-assisted relay communication networks is inevitably determined by the energy consumption of the UAVs. Similar to [12,14,29–32], our concern is the energy consumed by the UAV. The power consumption of UAV consists of two primary categories: propulsion-related power consumption P_{uav} and communication-related power consumption P_c. That is written as

$$P_{total} = P_{uav} + P_c \tag{12}$$

In this study, the completion time T is discretized into N sub-slots with width δ_t which is selected to be suitably tiny. It is assumed that during each sub-slot n, the UAV is in a state of quasi-static equilibrium and maintains a constant cruise velocity with negligible acceleration/deceleration [12,35–37]. Since the UAV is supposed to fly at a fixed altitude H, the propulsion power consumption of a rotary-wing UAV mainly depends on the flying speed [14]. As given in [31], the UAV propulsion power is represented as

$$P_{uav}[n] = c_0 + c_1\left(\sqrt{V[n]^4 + c_4} - V[n]^2\right)^{\frac{1}{2}} + c_2 V[n]^2 + c_3 V[n]^3 \tag{13}$$

where c_0 - c_4 are all constants related to the UAV hardware parameters. $V[n]$ is the scalar of the UAV velocity in the nth sub-slot, represented as $V[n] = \|\boldsymbol{V}[n]\|$, where $\boldsymbol{V}[n]$ is the velocity vector of the UAV horizontal flight in the nth sub-slot, and the relationship between the UAV position and velocity is expressed as

$$\boldsymbol{W}[n+1] = \boldsymbol{W}[n] + \boldsymbol{V}[n]\delta_t, \forall n \in (1, 2, \cdots, N-1) \tag{14}$$

It is worth noting that when substituting $V[n] = 0$ into formula (12), the propulsion power of the UAV when hovering can be calculated.

In sub-slot n, UAV serves a near-end user and a far-end user simultaneous, so, the communication power of UAV in sub-slot n is

$$P_c[n] = P_c{}^{near} + P_c{}^{far} \tag{15}$$

4.2. Optimization problem formulation

The energy efficiency is defined as the ratio of the sum of the data transmitted to the sum of the energy consumption in all sub-slots within T. Since the width of each sub-slot is equal, the energy efficiency can be defined as the ratio of the total system communication rate to the total power consumption, that is

$$EE = \frac{R_{total}}{P_{total}} \tag{16}$$

Combining formula (11), the total rate achieved by user m within T can be obtained as

$$R_{total}^m = \sum_{n=1}^{N} R_n^m \tag{17}$$

The total achievable rate of the system is determined by the cumulative achievable rates of all individual users, which can be written as

$$R_{total} = \sum_{m=1}^{M} R_{total}^m \tag{18}$$

Considering the QoS requirements of users, assuming that the total data amount achieved of user m is not less than the minimum data amount Th_m within T, i.e., $\sum_{n=1}^{N} \delta_t R_n^m \geq Th_m$. Since the width of each sub-slot is equal, the minimum total data amount constraint for user m within T can be expressed by the minimum total communication rate constraint. Therefore, the following constraint is applicable

$$R_{total}^m \geq R_{min}^m, \forall m \tag{19}$$

Combining formulas (12), (13) and (15), the power consumption P_{total} can be obtained as

$$\begin{aligned} P_{total} &= \sum_{n=1}^{N} (P_{uav}[n] + P_c[n]) \\ &= \sum_{n=1}^{N} [c_0 + c_1\left(\sqrt{V[n]^4 + c_4} - V[n]^2\right)^{\frac{1}{2}} \\ &\quad + c_2 V[n]^2 + c_3 V[n]^3 + P_n{}^{near} + P_c{}^{far}] \end{aligned} \tag{20}$$

According to formulas (16), (18) and (20), as well as relevant constraints, the energy efficiency optimization problem(P0) can be derived

by optimizing user scheduling F, power allocation P, and UAV trajectory $\boldsymbol{W}$, that is represented as

$$
\begin{aligned}
&\text{(P0)}: \max_{F,P,\mathbf{W}} EE \\
&\text{s.t. C1: } \mathbf{W}[1] = \mathbf{W}[N] \\
&\quad \text{C2: } \|\mathbf{V}[n]\| \le V_{\max}, \forall n \\
&\quad \text{C3: } \mathbf{W}[n+1] = \mathbf{W}[n] + \mathbf{V}[n]\delta t, \forall n \in \{1, 2, \dots, N-1\} \\
&\quad \text{C4: } \sum_{m=1}^{M} F(n, m, near) h_n^m > \sum_{m=1}^{M} F(n, m, far) h_n^m, \forall n \\
&\quad \text{C5: } P_n^{\text{near}} < P_n^{\text{far}}, \forall n \\
&\quad \text{C6: } P_n^{\text{near}} + P_n^{\text{far}} \le P_{\max}, \forall n \\
&\quad \text{C7: } R_{total}^{\text{m}} \ge R_{min}^{\text{m}}, \forall m \\
&\quad \text{C8: } \sum_{m=1}^{M} F(n, m, near) = 1, \forall n \\
&\quad \text{C9: } \sum_{m=1}^{M} F(n, m, far) = 1, \forall n \\
&\quad \text{C10: } F(n, m, near) = \{0, 1\}, \forall n \\
&\quad \text{C11: } F(n, m, far) = \{0, 1\}, \forall n
\end{aligned}
$$

where C1 indicates that the initial position of UAV is same as the final one. C2 limits the maximum speed of the UAV in the nth sub-slot. C3 gives the expression of position and speed of the UAV. C4 is the constraint on the channel quality of the near-end user and far-end user. C5 and C6 constrain UAV transmission power, where $P_{\max}$ in C6 is the maximum transmission power of UAV for sending data. C7 sets a minimum threshold of users' QoS requirements. C8-C11 ensure the correctness of user scheduling.

5. Proposed solutions

Due to the coupling of the three optimization variables in problem (P0) and the non-convex fractional form of the optimization objective, as well as the non-convex and mixed-integer programming constraints, it is not possible to directly solve problem (P0). To address this issue, problem (P0) is decomposed into three mutually independent subproblems: user scheduling F, power allocation P, and UAV trajectory planning $\boldsymbol{W}$, which are solved separately. The block coordinate descent method is then used to iteratively solve the three subproblems. In each iteration, only one subproblem is focused on. Using j to express the iteration number of the block coordinate descent method, F^j, P^j, and $\boldsymbol{W}^j$ represent the results of user scheduling, power allocation, and trajectory planning in the j-th iteration of the block coordinate descent method, respectively. Through multiple iterations, the appropriate solutions are obtained.

5.1. User scheduling

In the j-th iteration of the block coordinate descent, the UAV trajectory is fixed as the initial trajectory or the UAV trajectory obtained in the previous block coordinate iteration $\boldsymbol{W}^{j-1}$, and the power allocation is fixed as the initial power allocation or the power allocation obtained in the previous block coordinate iteration P^{j-1}. Now, the EE is determined only by user scheduling, and maximizing EE can be achieved by optimizing F^j. Additionally, since P_{total} is only related to the UAV trajectory, if the UAV trajectory is known, the maximization of EE is transformed into the maximization of R_{total}. Then, the user scheduling problem (P1) is expressed as

$$
\begin{aligned}
&\text{(P1)}: \max_{F^j} R_{total} \\
&s.t.\ \text{C4},\ \text{C7},\ \text{C8} - \text{C11}
\end{aligned} \tag{21}
$$

Problem (P1) is a standard integer programming problem. By using integer programming tools, the user scheduling results F^j in the j-th iteration of the block coordinate descent is obtained.

5.2. Power allocation

By using F^j obtained in problem (P1), combined with the initial UAV trajectory or the trajectory planning results obtained in the previous block coordinate iteration W^{j-1}, the user scheduling and UAV trajectory in the j-th iteration of the block coordinate descent can be fixed. This allows for the formulation of an EE optimization problem with P^j as the only variable, where EE is maximized by optimizing P^j. Similar to the user scheduling problem, the maximization of EE in this case can be transformed into the maximization of R_{total}, resulting in the power allocation problem (P2) written as

$$
\begin{aligned}
&\text{(P2)}: \max_{P^j} R_{total} \\
&s.t.\ \text{C5} - \text{C7}
\end{aligned} \tag{22}
$$

When the UAV transmission power is limited and considering the NOMA decoding rules, the power allocation of different users in the same sub-slots mutually influences each other. Additionally, the users' QoS requirements lead to the coupling of power allocations for the same user in different sub-slot. Therefore, problem (P2) is non-convex and cannot be solved directly. Specifically, when user m is served as a far-end user in the nth sub-slot, the term $R_{n_far_2}^m$ in formula (8) is non-convex and needs a convex approximation.

Since the system power consumption does not include the power consumption caused by the UAV sending information, and to facilitate the handling of the non-convex term in the rate expression of the far-end user, we assume that in each sub-slot, the UAV communicates with ground users using maximum transmission power $P_{\max}$, i.e., $P_n^{near} + P_n^{far} = P_{\max}$. Here, h_n^{far} and h_n^{near} are the channel power gains of the far-end and near-end users served by the UAV in the nth sub-slot. By reconfiguring formula (8), we obtain

$$
\begin{aligned}
R_{n_far_2}^m = &B\log_2\left(P_{max} h_n^{far} + \sigma^2\right) \\
&- B\log_2\left(P_n^{near} h_n^{far} + \sigma^2\right)
\end{aligned} \tag{23}
$$

formula (24) remains in a non-convex form, and the SCA technique is used to linearize it and then iteratively solve it. By performing a first-order Taylor expansion on the right side of equation (24) to find its global lower bound function, it is expressed as

$$
\begin{aligned}
R_{n_far_2}^m \ge\ & B\log_2\left(\sigma^2 + P_{max} h_n^{far}\right) \\
& - B\log_2\left(P_n^{near^{(l)}} h_n^{far} + \sigma^2\right) - \frac{B h_n^{far}\left(P_n^{near} - P_n^{near^{(l)}}\right)}{\ln 2\left(P_n^{near^{(l)}} h_n^{far} + \sigma^2\right)} \\
\triangleq\ & \tilde{R}_{n_far_2}^m
\end{aligned} \tag{24}
$$

where $P_n^{near^{(l)}}$ refers to the power allocated to the near-end user in the nth sub-slot obtained from the previous SCA iteration, and $\tilde{R}_{n_far_2}^m$ is a convex function. Therefore, for problem (P2), when user m is served as a far-end user, the communication rate is calculated by $\tilde{R}_{n_far_2}^m$,and then obtain new $\tilde{R}_{total}$ and $\tilde{R}_{total}^m$, leading to a new power allocation problem (P2.1). That is

$$
\begin{aligned}
&\text{(P2.1)}: \max_{P^j} \tilde{R}_{total} \\
&s.t.\ P_n^{near} < \frac{P_{max}}{2},\ \forall n \\
&\quad \tilde{R}_{total}^m \ge R_{min}^m,\ \forall m
\end{aligned} \tag{25}
$$

Problem (P2.1) is a standard convex optimization problem that can be directly solved by using convex optimization tools such as CVX. Then, we solve problem (P2) by using CVX to solve problem (P2.1), ultimately obtaining the power allocation result P^j. The steps to solve problem (P2)

Algorithm 1 Power allocation algorithm.

input: F^j, $\boldsymbol{W}^{j-1}$, SCA iterations s, maximum iterations $s_{\max}$, accuracy Φ
repeat
  update the objective function value $R^s_{total} \rightarrow R^{s-1}_{total}$;
  insert P^{s-1} into problem P2.1, then using CVX to solve problem (P2.1), obtain P^s;
  insert P^s into formula (23), then obtain R^s_{total};
  update variable $P^s \rightarrow P^{s-1}$;
until $\left|R^s_{total} - R^{s-1}_{total}\right| < \Phi$ or $s > s_{\max}$
output: P^j

are summarized in Algorithm 1. Since the complexity of Algorithm 1 primarily comes from iteratively solving the convex optimization subproblem (P2.1), the complexity of Algorithm 1 can be expressed as $\mathcal{O}(N^3 \cdot s_{\max})$, where N is the dimension of the power allocation variable P and $s_{\max}$ denotes the number of SCA iterations.

5.3. Trajectory planning

After obtaining user scheduling F^j and power allocation P^j, the only variable to be optimized is W^j. So, we have the optimization problem (P3). It is expressed as

$$\begin{aligned}&(\text{P3}): \max_{\boldsymbol{W}^j} EE\\ &s.t.\ \text{C1}-\text{C4},\ \text{C7}\end{aligned} \tag{26}$$

The trajectory of the UAV not only affects the channel quality of the users and the system's total rate R_{total} , but also directly determines the systems power consumption P_{total}. Therefore, problem (P3) is a fractional programming problem that can be transformed into a more easily solvable non-fractional programming problem using the Dinkelbach algorithm, and then traditional optimization techniques are used to solve the transformed problem. Specifically, using i to express the Dinkelbach iteration number, it takes the following form

$$EE^{i-1} = \frac{R_{total}\left(F^j, P^j, \boldsymbol{W}^{i-1}\right)}{P_{total}\left(\boldsymbol{W}^{i-1}\right)} \tag{27}$$

Through the Dinkelbach algorithm, problem (P3) can be transformed into problem (P3.1), which is expressed as

$$\begin{aligned}&(\text{P3.1}): \max_{\boldsymbol{W}^j} R_{total} - EE^{i-1}P_{total}\\ &s.t.\ \text{C1}-\text{C4},\ \text{C7}\end{aligned} \tag{28}$$

The position of UAV in each sub-slot affects channel quality of users, thereby affecting the communication rate that users can achieve in each sub-slot. However, due to the specific QoS requirements of users, the communication rates of the same user in different sub-slots are interdependent. In addition, the position of UAV in each sub-slot also affects the total system power consumption, this coupling relationship leads to the non-convexity of problem (P3.1), making it impossible to solve directly using conventional convex optimization methods. In this case, the SCA technique is used to linearize the non-convex problem and iteratively solve it.

Firstly, deal with the non-convex expression and constraints regarding the communication rate in problem (P3.1), namely the total rate *Rtotal* in the objective function and constraint C7, specifically including formulas (3), (7), and (8). For formula (3), we perform a first-order Taylor expansion on the right side of the equation to obtain its global lower bound function. That is written as

$$\begin{aligned}R_{n_1} &= B_m \log_2\left(1 + \frac{P_{BS}h_n^{uav}(\boldsymbol{W}[n])}{\sigma^2}\right)\\ &\geq B_m \log_2\left(1 + \frac{P_{BS}h_n^{uav}(\boldsymbol{W}[n]^{(l)})}{\sigma^2}\right)\\ &\quad - \frac{B_m b_0 P_{BS}\log_2(e)\left(\|\boldsymbol{W}[n]-BS\|^2 - \left\|\boldsymbol{W}[n]^{(l)}-BS\right\|^2\right)}{\sigma^2\left(\left\|\boldsymbol{W}[n]^{(l)}-BS\right\|^2 + H^2\right)\left(\left\|\boldsymbol{W}[n]^{(l)}-BS\right\|^2 + H^2 + \frac{b_0 P_{BS}}{\sigma^2}\right)}\\ &\triangleq \bar{R}_{n_1}\end{aligned} \tag{29}$$

where $\boldsymbol{W}[n]^{(l)}$ is the position of UAV in the nth sub-slot obtained from the previous SCA iteration. The global lower bound functions for formulas (7) and (8) are respectively expressed as

$$\begin{aligned}R^m_{n_near_2} &= B\log_2\left(1 + \frac{P_n^{near}h_n^{near}(\boldsymbol{W}[n])}{\sigma^2}\right)\\ &\geq B\log_2\left(1 + \frac{P_n^{near}b_0}{\sigma^2\left(\left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2 + H^2\right)}\right)\\ &\quad - \frac{Bb_0P_n^{near}\log_2(e)\left(\|\boldsymbol{W}[n]-U_m\|^2 - \left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2\right)}{\sigma^2\left(\left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2 + H^2\right)\left(\left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2 + H^2 + \frac{b_0P_n^{near}}{\sigma^2}\right)}\\ &\triangleq \bar{R}^m_{n_near_2}\end{aligned} \tag{30}$$

$$\begin{aligned}R^m_{n_far_2} &= B\log_2\left(1 + \frac{P_n^{far}h_n^{far}(\boldsymbol{W}[n])}{P_n^{near}h_n^{far}(\boldsymbol{W}[n]) + \sigma^2}\right)\\ &\geq B\log_2\left(1 + \frac{P_n^{far}b_0}{P_n^{near}b_0 + \sigma^2\left(\left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2 + H^2\right)}\right)\\ &\quad - \frac{BP_n^{far}\sigma^2b_0\log_2(e)}{P_n^{near}b_0 + \sigma^2\left(\left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2 + H^2\right) + P_n^{far}b_0}\\ &\quad \times \frac{\|\boldsymbol{W}[n]-U_m\|^2 - \left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2}{P_n^{near}b_0 + \sigma^2\left(\left\|\boldsymbol{W}[n]^{(l)}-U_m\right\|^2 + H^2\right)}\\ &\triangleq \bar{R}^m_{n_far_2}\end{aligned} \tag{31}$$

Using $\bar{R}_{n_1}$, $\bar{R}^m_{n_near_2}$ and $\bar{R}^m_{n_far_2}$ to calculate the communication rate of BS-UAV link, the near-end user and the far-end user in the UAV-user link in the nth sub-slot respectively, then we obtain new $\bar{R}^m_{total}$ and $\bar{R}_{total}$, and reconstruct problem (P3.1) to get a new problem (P3.2). That is

$$\begin{aligned}&(\text{P3.2}): \max_{\boldsymbol{W}^j} \bar{R}_{total} - EE^{i-1}P_{total}\\ &s.t.\ \text{C1}-\text{C4}\\ &\bar{R}^m_{total} \geq R^m_{min}, \forall m\end{aligned} \tag{32}$$

Algorithm 2 Trajectory planning algorithm.

input: F^j, P^j, Dinkelbach iterations i, SCA iterations k, maximum iterations $i_{\max}$, $k_{\max}$, accuracy Φ, objective function value sub of problem (P3.1)
repeat
  update EE, $EE^i \rightarrow EE^{i-1}$;
  repeat
    update sub, $sub^k \rightarrow sub^{k-1}$;
    insert EE^{i-1}, $\boldsymbol{W}^{k-1}$ into formula (36), using CVX to solve problem (P3.3), obtain $\boldsymbol{W}^k$;
    insert EE^{i-1}, $\boldsymbol{W}^k$ into formula (36), obtain sub^k;
    update variable, $\boldsymbol{W}^k \rightarrow \boldsymbol{W}^{k-1}$;
  until $|sub^k - sub^{k-1}| < \Phi$ or $k > k_{\max}$
  update variable $\boldsymbol{W}^{k-1} \rightarrow \boldsymbol{W}^i$, insert $\boldsymbol{W}^i$ into formula (15), obtain EE^i;
until $|EE^i - EE^{i-1}| < \Phi$ or $i > i_{\max}$
output: $\boldsymbol{W}^j$

Algorithm 3 Global optimization algorithm.

input: initialize $\boldsymbol{W}^0$, P^0, block coordinate iterations j, maximum iterations $j_{\max}$, accuracy Φ
repeat
  update $EE\ EE^j \rightarrow EE^{j-1}$;
  Given $\boldsymbol{W}^{j-1}$ and P^{j-1}, using integer programming tool to solve problem (P1), obtain F^j;
  Given $\boldsymbol{W}^{j-1}$ and F^j, using Algorithm 1 to solve problem (P2), obtain P^j;
  Given F^j and P^j, using Algorithm 2 to solve problem (P3), obtain $\boldsymbol{W}^j$;
  insert F^j, P^j, $\boldsymbol{W}^j$ into formula (16), obtain EE^j;
  update variable $P^j \rightarrow P^{j-1}$, $\boldsymbol{W}^j \rightarrow \boldsymbol{W}^{j-1}$;
until $|EE^j - EE^{j-1}| < \Phi$ or $j > j_{\max}$
output: F, P and $\boldsymbol{W}$ that maximizes EE

Next, it is necessary to deal with the non-convex expressions related to the P_{total}, namely linearizing formula (13). By introducing a relaxation variable $u[n]$ to deal with the non-convex terms in formula (13) and with $u[n] \geq \left(\sqrt{V[n]^4 + c_4} - V[n]^2\right)^{\frac{1}{2}}$, we have the following inequality

$$\frac{c_4}{u[n]^2} \leq u[n]^2 + 2\|\boldsymbol{V}[n]\|^2 \tag{33}$$

where the right side of formula (33) is still a non-convex expression about $\boldsymbol{W}[n]$, but is convex about $u[n]$ and $\boldsymbol{V}[n]$. By performing a first-order Taylor expansion on them and let $low[n]$ represent its lower bound function, we get

$$\begin{aligned}\frac{c_4}{u[n]^2} \leq low[n] &= (u[n]^{(l)})^2 + 2u[n]^{(l)}(u[n] - u[n]^{(l)}) \\ &+ 2\left\|V[n]^{(l)}\right\|^2 + 4(V[n]^{(l)})(V[n] - V[n]^{(l)})\end{aligned} \tag{34}$$

where $u[n]^{(l)}$ is the value of $u[n]$ in the previous SCA iteration, and $V[n]^{(l)}$ is the value of $V[n]$ in the previous SCA iteration. By reconstructing formula (19), we obtain the new system power consumption as

$$\bar{P}_{total} = \sum_{n}^{N} c_0 + c_1 u[n] + c_2 V[n]^2 + c_3 V[n]^3 \tag{35}$$

Using $\bar{P}_{total}$, we transform the optimization problem (P3.2) into problem (P3.3). That is

$$\begin{aligned}(\text{P3.3}):\ &\max_{\boldsymbol{W}^j} \bar{R}_{total} - EE^{i-1}\bar{P}_{total} \\ &s.t.\ \text{C1} - \text{C4} \\ &\bar{R}^m_{total} \geq R^m_{min}, \forall m \\ &(35),\ \forall n\end{aligned} \tag{36}$$

Problem (P3.3) is a standard convex optimization problem that can be solved using the convex optimization tool CVX. Then, we perform SCA iterations on problem (P3.3) to solve it, thereby addressing the optimization problem (P3.1). Finally, we apply Dinkelbach iterations on problem (P3.1) to solve problem (P3), ultimately obtaining the trajectory planning result $\boldsymbol{W}^j$. The trajectory planning solution steps can be summarized as Algorithm 2. The overall trajectory planning algorithm to solve problem (P3) is given in Algorithm 2. By analyzing its nested-loop structure, the inner-layer complexity is dominated by solving the standard convex optimization problem (P3.3). The complexity of Algorithm 2 is $\mathcal{O}(N^3 \cdot k_{\max} \cdot i_{\max})$, where N is the dimension of the UAV trajectory variable W, $i_{\max}$ and $k_{\max}$ are the iteration counts for the Dinkelbach loop and SCA loop.

5.4. Global optimization

Based on the solution methods for subproblems (P1), (P2) and (P3), the block coordinate descent method is used to iteratively solve the three subproblems alternately, obtaining a feasible solution for the original optimization problem(P0). For detailed steps, refer to Algorithm 3. Algorithm 3 covers the three subproblems (P1), (P2) and (P3). Problem (P1) is solved using an integer programming tool, while problem (P2) and problem (P3) are transformed into standard convex problems and solved by using Algorithms 1 and 2, respectively. The objective value of the original optimization problem(P0) in the $(j+1)$th iteration is not less than the objective value obtained in the jth iteration, indicating a non-decreasing trend with iterations. Additionally, due to the finite communication resources, including the limited number of sub-slots for user scheduling and transmission power, the objective value has an upper bound, i.e., EE is a finite value. Therefore, the convergence of Algorithm 3 is guaranteed. The complexity of Algorithm 3 is mainly derived from Algorithms 1 and 2, the overall complexity of Algorithm 3 can be expressed as $\mathcal{O}\left((N^3 \cdot s_{\max} + N^3 \cdot k_{\max} \cdot i_{\max}) \cdot j_{\max}\right)$, where $i_{\max}$ denotes the maximum number of Dinkelbach iterations in Algorithm 2 and $j_{\max}$ is the maximum number of block coordinate iterations in Algorithm 3.

Table 2
List of simulation parameters.

Parameter	Value
Noise power (σ^2)	-50 dB
UAV's maximum flight speed ($V_{\max}$)	40 m/s
UAV's flight altitude (H)	50 m
UAV's maximum transmission power ($P_{\max}$)	1 mW
Number of Ground Users	6
BS's transmission power (P_{BS})	10 W
Task completion time (T)	60 s
User's QoS Requirements ($R^m_{\min}$)	5bit/s/Hz
Channel bandwidth	1 MHz
Number of sub-slots (N)	120
Channel power gain (b_0)	-30 dB
Initial transmission power for the near-end user (P_n^{near})	0.4 mW
Initial transmission power for the far-end user (P_n^{far})	0.6 mW

6. Simulation results and analysis

To verify the effectiveness of the proposed scheme, we compare it with NOMA and fixed trajectory in [28], named scheme2, where communication is conducted using NOMA and the UAV flies along a circular trajectory with an optimized radius, OMA and trajectory planning in [20], named scheme3, in which communication is done using OMA technology and trajectory planning is performed, OMA and fixed trajectory in [15], named scheme4, which uses OMA with an optimized circular trajectory. Similar to [31], the simulation parameters are set as follows. The BS position is $BS = (0, 0)$, the BS transmission power is $P_{BS} = 10$ W, there are 6 ground users, and the exact coordinates of the six users can be found in Fig. 4. The parameters related to the UAV hardware in its propulsion power consumption model are set as $c_0 = 30$, $c_1 = 70.7$, $c_2 = 0.15$, $c_3 = 0.064$ and $c_4 = 0.3906$. The minimum QoS of users is set to 5, 20 or 25 bit/s/Hz. Additionally, given the importance of the number of sub-slots to the simulation, and considering its compatibility with the mission time (T) as well as its impact on the iteration efficiency of the optimization algorithm, the number of sub-slots is ultimately set to N = 120. Other parameters of the networks are listed in Table 2.

Since assigning a high-quality initial value to the variables in BCD algorithm is crucial for accelerating convergence and final performance, a well-designed starting point ensures that the subsequent SCA process, which generates the convex problems for the CVX solver, begins in a feasible and reasonable region. Our initialization strategy for both initial power and trajectory is detailed as follows.

For the initial power (P^0), the choice is analytically derived to satisfy the fundamental constraints of the NOMA system from the outset, specifically the power ordering rule (C5) and the total power limit (C6).

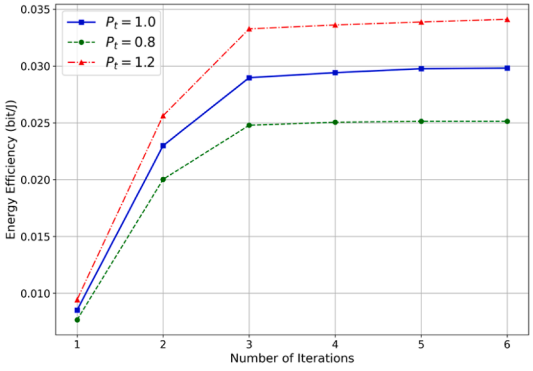


Fig. 2. Convergence performance under different transmission power..

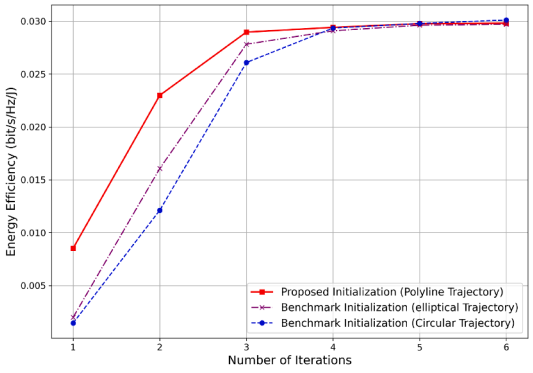


Fig. 3. Convergence performance under different initial trajectories..

Given $P_{\max} = 1$ mW, we set the initial power for the near-end user to 0.4 mW and for the far-end user to 0.6 mW. This ensures the algorithm starts from a valid state. Fig. 2 illustrates the impact of power-related parameters on the algorithm's convergence with different initial points derived from different maximum transmission power (P_t) values. The results show that while the algorithm remains robust, the choice of these parameters significantly influences the convergence path and final energy efficiency.

To test the influence of UAV's initial trajectory (W^0) on system convergence performance, we propose a pre-calculated, non-symmetric polyline path designed to provide a rough coverage of all ground users and avoid the numerical instabilities associated with perfectly geometric paths. To empirically validate the importance of this design, we compared its convergence performance against two benchmarks: a circular and an elliptical trajectory. As shown in Fig. 3, our proposed initialization exhibits the fastest convergence speed, reaching a high-performance state within just three iterations, significantly outperforming the benchmarks. This experiment confirms that a well-designed initial trajectory is critical for accelerating the algorithm's convergence. The optimized initial trajectory of the UAV that has been verified by the above results is employed for both shorter flight distance and better channel quality, and can be converged to a stable value faster by the block coordinate descent algorithm.

Fig. 4 shows the UAV trajectory under different users' QoS requirements in the proposed method. According to the channel gain between the UAV and users in formula (2), an increase in QoS means that users require a higher communication rate, and a higher rate needs better channel gain. The channel gain is inversely proportional to the square of the distance between the UAV and the user. Therefore, the UAV must shorten its distance to users, resulting in the trajectory adjusting from concentrated coverage of the central area (blue curve, which only needs to serve nearby users to meet the minimum QoS, such as 5 bit/s/Hz) to a distributed trajectory surrounding all users (red and green curves, with

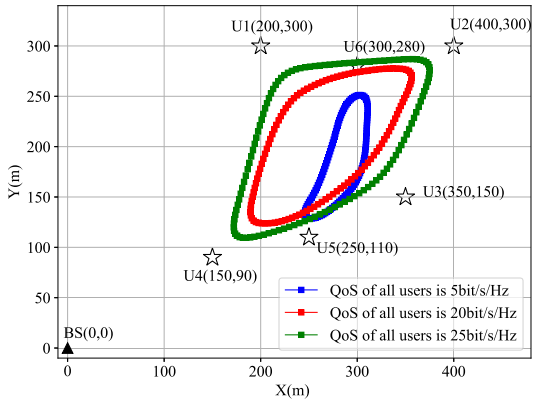


Fig. 4. UAV trajectories of the Proposed Scheme under different users' QoS requirements..

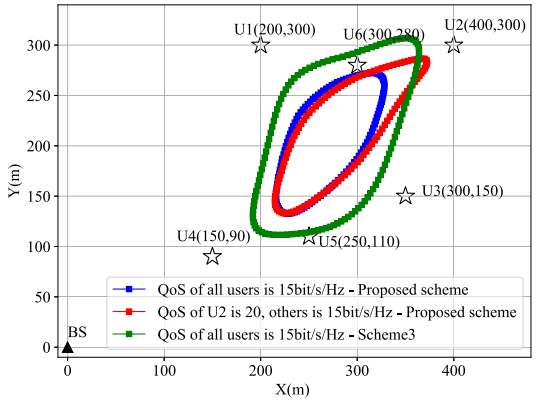


Fig. 5. UAV trajectory under different multiple access method and different users' QoS requirement.

QoS requirements of 20 bit/s/Hz and 25 bit/s/Hz respectively). The UAV needs to approach each distant user one by one to obtain better channel gain. It is worth noting that the longer trajectory directly results in an increase in the UAV's propulsion power consumption according to formula (13), which has a significant impact on the improvement of energy efficiency. On the one hand, the increased flight distance raises the cumulative speed value over the total flight time, leading to a significant increase in power consumption contributed by the $V[n]^2$ and $V[n]^3$ terms. On the other hand, frequent adjustments to the flight direction (such as the multiple inflection points in the green curve) cause fluctuations in the instantaneous speed, further increasing the additional power consumption of non-uniform flight.

Fig. 5 compares UAV trajectory when using NOMA and OMA respectively, as well as the impact of changes in individual users' QoS requirements on the trajectory under NOMA. From Fig. 5, it can be observed that when facing the same users' QoS requirements, NOMA can satisfy the users' QoS requirements with a relatively shorter flight distance, while under OMA, the UAV trajectory needs to be closer to the users to meet the users' QoS requirements. The reason is that NOMA adopts power domain multiplexing, allowing it to serve one near-end user and one far-end user simultaneously in each sub-slot according to formulas (C8-C9). By using successive interference cancellation (SIC) technology to eliminate interference, there is no need to allocate independent resource blocks to each user. Therefore, the UAV does not need to approach all users one by one and only needs to be in an intermediate position that can cover one near-end and one far-end user simultaneously to meet the rate requirements (such as the blue curve). However, OMA allows only one user to occupy the time resource at a

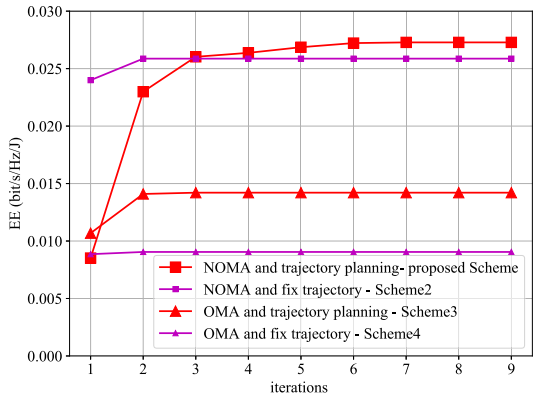


Fig. 6. Comparison of energy efficiency with iterations under different schemes.

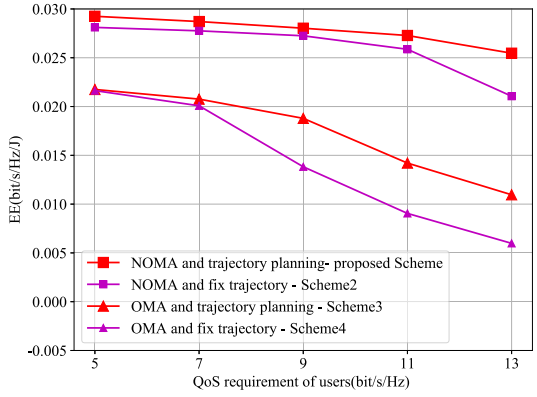


Fig. 7. Comparison of energy efficiency with users' QoS requirement of different schemes.

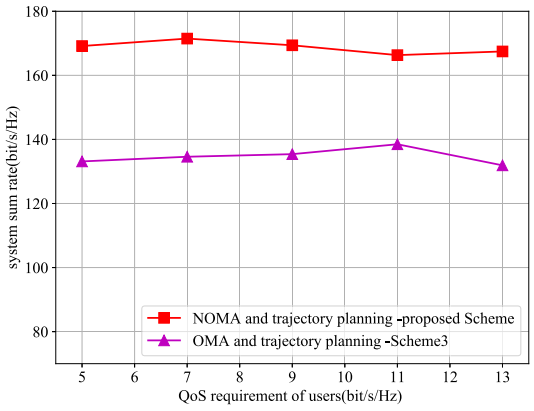


Fig. 8. The system total rate with users' QoS requirement under different schemes.

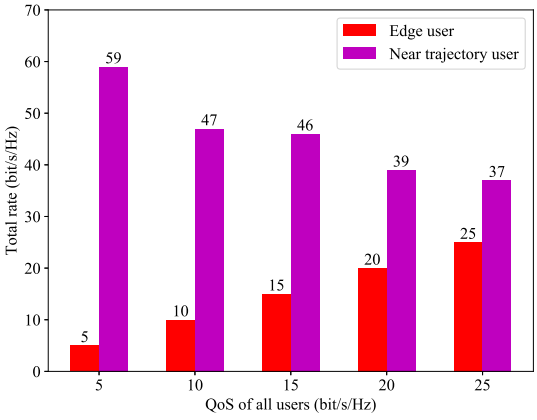


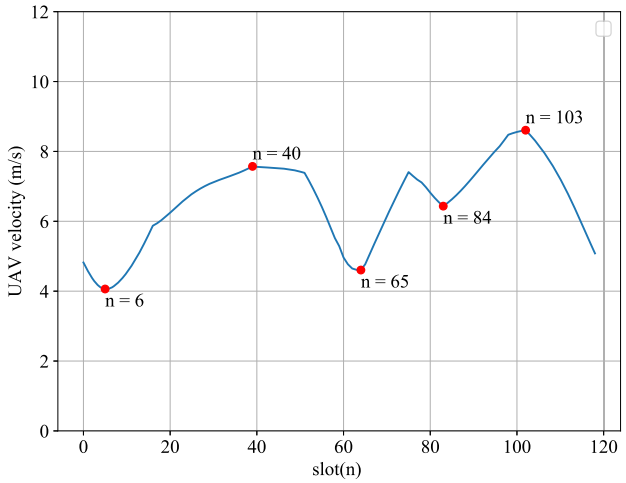
Fig. 9. The impact of user location on the total rate of the users with different QoS.

time, resulting in lower spectrum utilization than NOMA. To meet the QoS of all users, the UAV must fly to the vicinity of each user in sequence (such as the green curve, which has a larger trajectory coverage range), leading to a significant increase in flight distance. In addition, under the NOMA scheme, when the QoS of User 2 (U2) increases from 15 bit/s/Hz to 20 bit/s/Hz, the UAV trajectory shifts toward User 2 (such as the red curve). This is because the UAV needs to provide better channel quality to meet U2's QoS requirements.

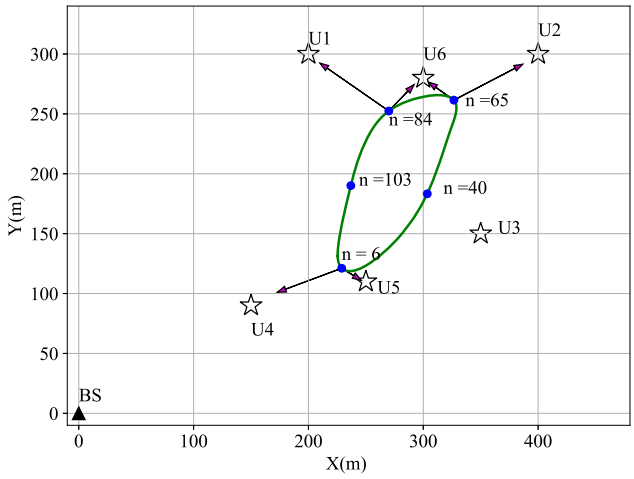
Fig. 6 shows the change in *EE* with the iterative process of Algorithm 3 and the comparison with other schemes. It can be seen from Fig. 6 that the *EE* eventually converges when using block coordinate iteration. Under the premise of using the same multiple access method for communication, compared with UAV flying on a fixed circular trajectory, optimizing UAV trajectories can improve *EE* by 3.8 %. The reason is that the optimized trajectory can adapt to changes in user positions and users' QoS requirements in a more flexible manner, providing better channel quality to users with lower P_{total} to achieve a higher R_{total}, leading to a higher *EE*. We also observed that in the initial iteration stage, the energy efficiency of our scheme is not the highest, but after a certain number of iterations, it reaches the highest level. That is because the proposed scheme adopts the block coordinate descent method, which requires multiple iterations to gradually approach the optimal solution. In the first two iterations, only some variables are optimized and a globally optimal combination has not yet been formed, while the other schemes adopt simpler optimization strategies (only optimizing power with a fixed trajectory). Although they can quickly achieve a relatively high initial energy efficiency, their global optimality is not as good as ours. Additionally, under the premise of UAV trajectory planning, the *EE* using NOMA is 92.9 % higher than using OMA, determined by NOMA advantage in improving communication rates relative to OMA[22].

The two schemes in Fig. 8 aim to highlight the impact of different multiple access methods on the system total rate. For a more targeted comparison focusing only on the impact of multiple access methods on the system total rate, both schemes optimize the UAV trajectory. From Fig. 8, it can be seen that when facing the same users' QoS requirements, compared to OMA, using NOMA communication can increase the system sum communication rate by at least 20.2 % which also confirms the advantage of NOMA over OMA in improving communication rates. A higher system total rate is also more conducive to achieving higher *EE*.

Fig. 7 shows the changing trend of *EE* under different schemes as the QoS requirements of all users gradually increase. From Fig. 7, it can be seen that *EE* under the four schemes continuously decreases. This is because when the QoS requirements of all users are low, the UAV satisfies the users' QoS requirements with a relatively short flight distance, corresponding to the trajectory shown by the blue curve in Fig. 4. A shorter flight distance corresponds to lower system power consumption and higher *EE*. However, as the users' QoS requirements increase, the trajectory needs to be closer to the users to provide better channel quality to meet the requirements, as shown by the red and green curves in Fig. 4, leading to an increase in UAV flight distance. A longer flight distance corresponds to higher system power consumption and lower *EE*. Nevertheless, the proposed scheme in this paper, combining NOMA technology with trajectory planning, can achieve higher *EE* compared to other schemes when the users' QoS requirements are the same. In addition, when the QoS increases from 5 bit/s/Hz to 20 bit/s/Hz, the EE loss rate of the proposed scheme is the lowest among all schemes. This is because trajectory optimization dynamically compensates for channel losses, and the power multiplexing of NOMA reduces resource waste. These two factors together reduce the sensitivity to QoS improvement,



(a) UAV flight velocity at different slots



(b) UAV position and user scheduling at different slots

Fig. 10. The relationship between UAV speed, position and user scheduling.

indicating that the proposed scheme has high robustness under different QoS requirements. To address the limited processing capabilities inherent to emergency situations, our algorithm leverages a computationally efficient convex optimization tool to expedite the convergence of Algorithm 1. Consequently, this approach is better aligned with the practical demands of emergency applications, and the resulting suboptimality is considered an acceptable trade-off.

Fig. 9 compares the total rates achievable by edge users and near-trajectory users. To simplify the analysis, User 1 is considered as the edge user and User 5 as the near-trajectory user in Fig. 9, with their respective total rates indicated. The near-trajectory user has better channel conditions compared to the edge user, allowing them to achieve higher communication rates. In order to maximize *EE* and meet the QoS requirements of each user, it is necessary to allocate as many communication resources as possible to the near-trajectory user. Therefore, the total rate achievable by the near-trajectory user is much higher than their QoS requirements, as shown by the purple bar in Fig. 9. On the other hand, the edge user, facing the same communication resources, achieves lower communication rates due to their poorer channel conditions, which hinders achieving higher *EE*. Hence, the communication resources allocated to the edge user are just enough to meet their QoS requirements, as shown by the red bar in Fig. 9.

Fig. 10 illustrates the relationship between the velocity of the UAV and user scheduling over the entire task completion time T. As shown in Fig. 10(a), the UAV flight speed is lower near the 6th, 65th, and 84th sub-slots. Combining with Fig. 10(b), this is because the UAV is positioned optimally to serve User 4 and User 5 in the 6th sub-slot, User 2 and User 6 in the 65th sub-slot, and User 1 and User 6 in the 84th sub-slot, resulting in the UAV flying at a lower speed to provide more sub-slots and achieve a higher total rate. Near the 40th sub-slot, the UAV flight speed is higher because at that sub-slot, the UAV is positioned farther away from all users, resulting in poorer channel quality and hindering the achievement of higher system total rate and *EE*, hence the faster flight speed. Near the 103th sub-slot, the UAV reaches the fastest flight speed over the entire duration T, this is because at this sub-slot, the UAV is positioned at the location with the poorest communication channel quality with all users in T, resulting in the fastest flight speed. In conclusion, by adjusting the UAV flight path and speed, it is possible to achieve the lowest possible system power consumption and provide the best possible channel quality, thereby achieving higher energy efficiency.

7. Conclusion and future work

This paper focuses on addressing the critical challenges in UAV-assisted real-time relay communication networks which are required to provide diverse communication QoS for a large number of users with few resources. So, we introduced NOMA technology into UAV-assisted real-time relay communication networks to improve the network capacity, and formulated an energy efficiency maximization problem under various users' QoS requirements and UAV's energy constraints to prolong the network lifetime. Considering the strong coupling among various factors and non-convexity of the optimization problem, we decomposed it into three interrelated sub-problems: user scheduling, transmission power allocation, and UAV trajectory planning and solve them iteratively by using integer programming, SCA, and the Dinkelbach transformation. Extensive simulation results validated the effectiveness of the proposed scheme, demonstrating significant energy efficiency improvements over the compared schemes. With the rapid development and wide application of UAV-assisted communication networks, they are confronted with more complex and uncertain environments. How to dynamically allocate the limited communication resources in more complicated scenarios, thereby achieving better communication performances while reducing energy consumption, is our goal to strive for it. Therefore, in future work, we will explore dynamic energy efficiency optimization methods. If the computing power and hardware resources of UAV-assisted communication networks permit, various machine learning methods are attempted to enhance the network performances. Meanwhile, we will adopt more comprehensive metrics to evaluate the network performance, such as Age of Information (AoI), fairness, etc., to meet the actual application requirements.

CRediT authorship contribution statement

Min Li: Writing – review & editing, Supervision, Project administration; **Yong Ku:** Writing – review & editing, Validation, Investigation; **Ligang Zhu:** Writing – review & editing, Validation; **Huajian Zhou:** Writing – original draft, Validation, Investigation; **Heng Wang:** Funding acquisition, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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